Student Dropout Prediction: A Machine Learning Analysis

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset through preprocessing, t-SNE visualization, clustering, supervised classification, model evaluation, and open-ended exploration. The engineered top-40 feature space is dominated by academic performance signals such as semester approval rates and grades. Unsupervised analysis shows that Enrolled students lie between Dropout and Graduate rather than forming a clean natural cluster. Logistic Regression is the strongest mandatory baseline. In open-ended exploration, validation-tuned class-bias adjustment for LR achieves the best held-out test macro F1 of 0.7228, improving the Enrolled decision boundary while remaining only borderline significant over plain LR.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository [1] contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters.

The report follows an evidence-first structure: construct an engineered top-40 feature space, use t-SNE and clustering to test whether outcomes form natural groups, then evaluate supervised decision boundaries. Because Enrolled students remain structurally intermediate between Dropout and Graduate, the open-ended section focuses on class-boundary refinement rather than only adding model complexity.

2 Mandatory Tasks

2.1 Data Preprocessing

The dataset contains 36 raw features that are categorized into four types: *nominal* (8 features), *binary* (8 features), *ordinal* (13 features), and *continuous* (7 features). It also includes a three-class target variable: `Dropout` (32.1%), `Enrolled` (18.0%), and `Graduate` (49.9%). We performed a comprehensive data preprocessing pipeline as outlined below.

Missing values and outliers. The dataset is clean with no missing values, as confirmed by a complete null-value check.

EDA for nominal features. We inspected the distribution and top categories of nominal features. Except for feature `Course`, most nominal features are heavily skewed and the top four categories can cover over 80% of the data. This informed our decision to apply out-of-fold (OOF) target encoding for these features, as one-hot encoding would lead to high dimensionality and sparsity.

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EDA for numerical correlations. We sorted numerical features by absolute Pearson correlation with the target and visualized the top 15 with $|r| \geq 0.09$ (Figure 1). Curricular-unit approval and grade features show the strongest signal, while economic and demographic variables provide weaker but still useful complementary information.

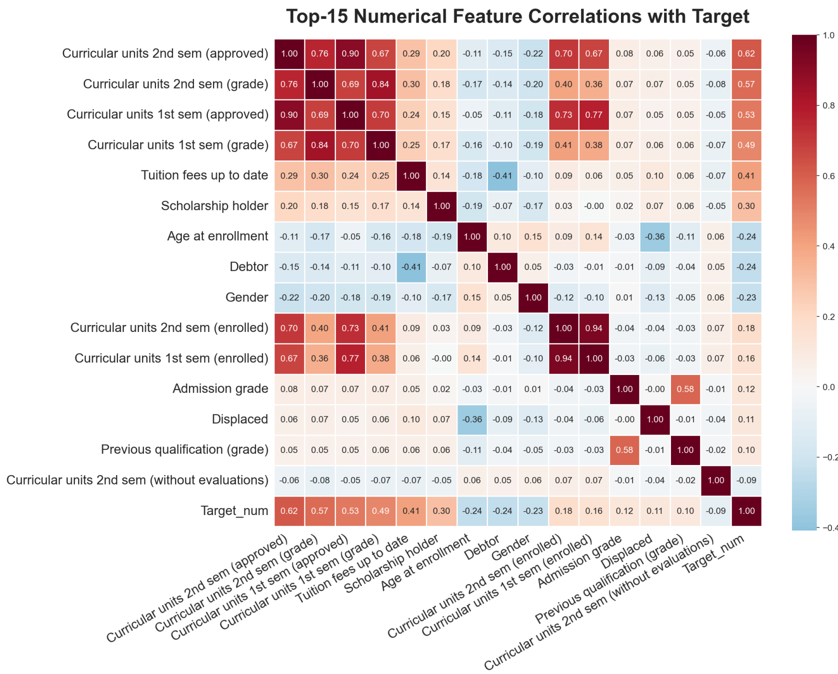

Figure 1: Top-15 feature correlations with the target variable.

EDA for important features. We visualized four key curricular-unit features as histograms with trendlines (Figure 2). Higher values are associated with Graduate, lower values with Dropout, and Enrolled students remain intermediate.

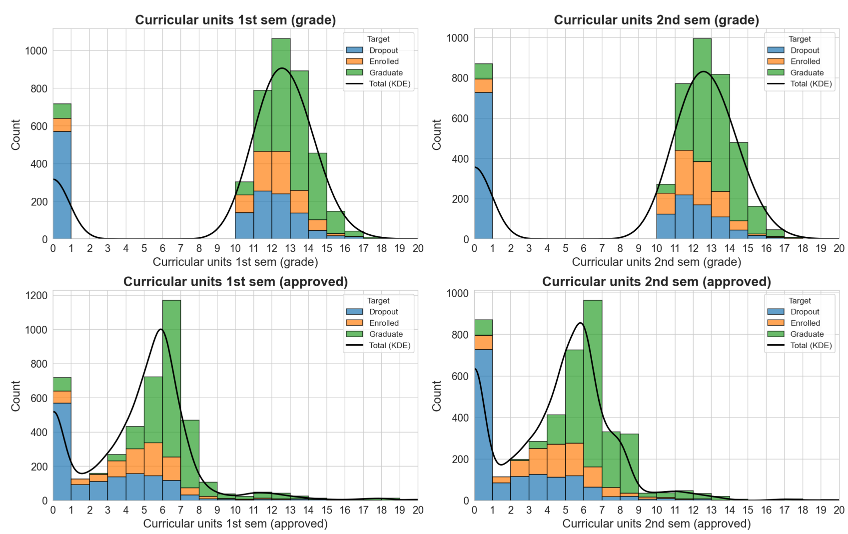

Figure 2: Distributions of top curricular unit features across target classes.

Feature engineering and selection. We created 25 domain-specific features from the 36 raw features, yielding 61 candidates.

Data splitting and target encoding. The dataset was split into stratified train/validation/test sets (70/15/15). OOF target encoding was fit on training folds for nominal features and then applied to validation and test sets, preventing leakage while capturing categorical signal.

Feature standardization, selection, and output. A Random Forest selected the top 40 target-encoded features by importance. Non-binary selected features were standardized using training-set statistics only; binary indicators remained 0/1, and the fitted scaler was reused for validation and test sets.

2.2 Data Visualization (t-SNE)

We applied t-SNE to the engineered feature dataset to project the high-dimensional feature space into lower dimensions for visualization.

A systematic pattern analysis across four perplexity values 10, 20, 30, and 50 highlights the algorithm's sensitivity to local versus global structure, as shown in Figure 3.

At a low perplexity ($p = 10$), the projection is severely fragmented. Higher perplexity values ($p = 20, 30, 50$) produce similar stable global structures and the relative positions of the three classes remain consistent. However, high perplexity tends to over-smooth the local structure and concentrate points into denser clusters. Generally, perplexity 30 achieves the optimal balance.

The 2D projection reveals that Graduate and Dropout students form relatively compact clusters, while Enrolled students scattered between both groups and mostly overlap with the Graduate cluster, indicating that Enrolled students do not form a distinct cluster but rather occupy an intermediate region in the feature space. This pattern is consistent across all perplexity values, confirming that Enrolled students have transitional academic status. The overlapping nature of the Enrolled class suggests that it may be more challenging to classify them accurately based on the engineered features or clustering alone.

Furthermore, extending the projection from 2D to 3D (Figure 4) provides additional spatial volume, further reducing the overlap between the distinct Graduate and Dropout clusters.

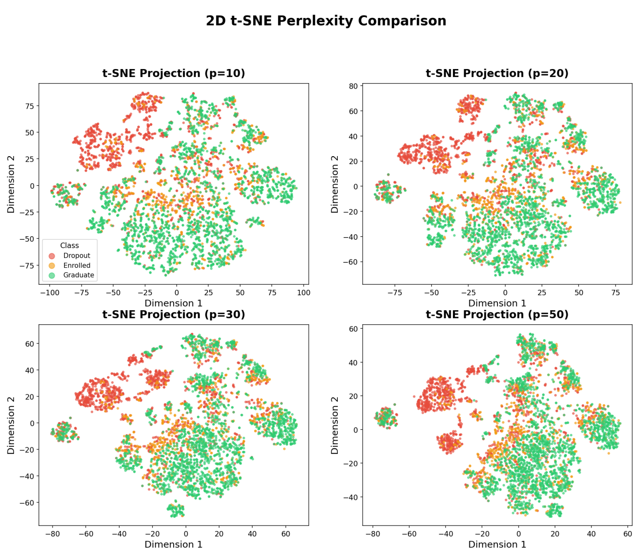

Figure 3: t-SNE 2D projections at perplexity 10, 20, 30, 50.

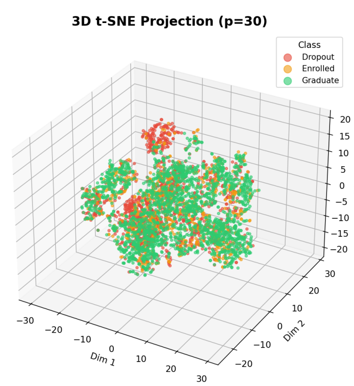

Figure 4: t-SNE 3D projection at perplexity 30.

2.3 Clustering Analysis

2.3.1 Algorithm Exploration and Comparison

We applied three clustering algorithms to the engineered feature dataset: K-Means, Agglomerative Hierarchical Clustering, and Gaussian Mixture Models (GMM). 2D t-SNE projections with perplexity 30 were used for visualization, which can be compared with the original t-SNE visualization (Figure 3) to assess how well the clusters align with the underlying structure (true label) of the data.

K-Means K-Value Exploration. We tested $K = 3, 4, \ldots, 19$. $K = 4$ achieved the best ARI of 0.2313; $K = 5$ achieved the best DB index of 2.2165; but other metrics favored $K = 3$ and monotonically worsened with larger K. The detailed metrics for $K = 3, 4, 5, 6$ are shown in Table 1.

Hierarchical Clustering. We first compared four linkage methods with $K = 3$ to identify the best-performing method, then explored K values for that method. The ARI results show that ward linkage significantly outperforms complete, average, and single linkages which produce degenerate partitions with ARI close to zero.

K-value exploration with Ward linkage showed similar patterns to K-Means: ARI peaked at K=3 (0.1332), Silhouette and DB index favored K=4, and the overall pattern of worsening metrics with larger K values. Detailed metrics for Ward linkage with $K = 3, 4, 5, 6$ are shown in Table 2.

Table 1: K-Means K-value exploration on engineered features.

K	Silhouette	CH Index	DB Index	ARI	NMI
3	0.1301	**848.80**	1.5689	**0.2722**	0.1198
4	**0.2313**	729.24	**2.0620**	0.1734	**0.1559**
5	0.2004	629.19	2.2165	0.1422	0.1486
6	0.1544	576.00	1.9510	0.1246	0.1306

Table 2: Hierarchical Clustering K-value exploration (Ward linkage).

K	Silhouette	CH Index	DB Index	ARI	NMI
3	0.2538	**772.45**	1.6391	**0.1332**	**0.1187**
4	**0.2650**	651.54	1.8610	0.1176	0.1113
5	0.2539	563.06	1.5257	0.1149	0.1098
6	0.1226	513.87	**1.9085**	0.0898	0.0932

GMM K-Value and Covariance Exploration. We tested $K = 3, 4, 5, 6$ combined with four covariance types (full, tied, diag, spherical). Although $K = 3$ with diag covariance achieved the best ARI 0.2502, $K = 4$ with tied covariance achieved a better Silhouette and close metric values. For consistency with the other algorithms, we selected $K = 4$ with tied covariance as the best GMM configuration.

Overall Algorithm Comparison: Table 3 summarizes all three algorithms with their optimal configurations. GMM (K=4, tied) leads in ARI (0.2270), NMI (0.1597) and DB Index (2.1498). However, the gap of the metrics among the three algorithms is minor, and overall performance is modest across all algorithms. This may suggest that the global clustering structure of the dataset is relatively strong and stable, leading to similar partitions across different algorithms, but the structural overlap between classes (especially Enrolled) limits the maximum achievable ARI and NMI.

Figure 6 shows the cluster assignments of the three algorithms and true labels on the 2D t-SNE projection. K-Means produces compact clusters but with significant label mixing, while the results of Hierarchical clustering and GMM show more mixing and overlap patterns between the clusters. Nevertheless, all three algorithms capture similar global structures that is highly consistent with the true labels, with one cluster dominated by Graduate students, one cluster dominated by Dropout students, and one cluster that is a mixture of Enrolled and other classes.

However, this observed consistency between the clustering results and the true labels may be caused by label leakage from the OOF target encoding step, which can create artificial correlations between the features and the target variable. We therefore designed a follow-up ablation experiment to test the impact of target encoding on clustering performance in the next section.

Table 3: Clustering algorithm performance comparison.

Algorithm	K	Silhouette	CH Index	DB Index	ARI	NMI
K-Means	4	0.2313	**729.24**	2.0620	0.1734	0.1559
Hierarchical (Ward)	4	**0.2650**	651.54	1.8610	0.1176	0.1113
GMM	4	0.1697	678.75	**2.1498**	**0.2270**	**0.1597**

2.3.2 Clustering Ablation: Impact of Target Encoding

To test the impact of OOF target encoding on clustering performance, we conducted an ablation experiment where we removed all target-encoded features and re-ran the clustering algorithms on the remaining 23 features (standardized numerical features and binary features).

The clustering performance slightly decreased across all algorithms, comfirming that target encoding contributed to the clustering structure. However, the overall patterns remained consistent: all three algorithms still produced similar global structures with one cluster dominated by Graduate

True labels

K-Means (K=4)

Hierarchical (K=4, ward)

GMM (K=4, tied)

Figure 5: Cluster visualization on 2D t-SNE projections.

students and one cluster dominated by Dropout students, while Enrolled students does not form a distinct cluster among the remaining features but instead overlap with the other two classes. This suggests that while target encoding enhances the clustering signal especially for Enrolled students, the underlying structure of the data still reflects the same class relationships.

2.3.3 Density-Based Clustering Exploration

DBSCAN and OPTICS were explored but failed: DBSCAN classified 100% of points as noise across all `eps` values (0.3-1.5), and OPTICS with xi-based extraction consistently returned only one cluster. Both failures stem from the curse of dimensionality — in 40 dimensions all pairwise distances become similar (coefficient of variation = 0.32), making density-based methods unable to identify dense regions.

2.4 Prediction: Training and Testing

The clustering results suggest that the three student outcomes are not naturally separated in the engineered feature space. We therefore train supervised classifiers on the top-40 engineered features to test whether label-informed decision boundaries can better distinguish Dropout, Enrolled, and Graduate students.

Data preparation. We used the engineered feature dataset (40 features after OOF target encoding and top-40 selection), split into train (3,096), validation (664), and test (664) sets. The class distribution in the training set is: Dropout 32.1%, Enrolled 17.9%, Graduate 49.9%.

Decision Tree (DT), Logistic Regression (LR), and RBF-SVM were selected because they represent complementary assumptions: non-linear splits, interpretable linear boundaries, and kernel-based nonlinear boundaries.

True labels | K-Means (K=4)

Hierarchical (K=4, ward) | GMM (K=4, tied)

Figure 6: Cluster visualization on 2D t-SNE projections without target-encoded features.

We trained each classifier using default configurations from `prediction.py`: Decision Tree (`max_depth=10, class_weight='balanced'`), Logistic Regression (`solver='saga', class_weight='balanced'`), and SVM with RBF kernel (`class_weight='balanced'`). All models were evaluated on training, validation, test, and entire sets.

Multi-set evaluation. Table 4 reports accuracy and macro F1 across all evaluation sets. DT has the largest train-test gap, indicating overfitting. LR has the strongest held-out test macro F1 among the mandatory baselines and remains stable across train, validation, and test splits. Default RBF-SVM improves substantially after the corrected scaling step, but LR remains the strongest mandatory baseline by held-out macro F1.

The per-class analysis shows that all models achieve the highest recall for Graduate and the lowest for Enrolled. This is consistent with the class imbalance and the transitional nature of enrolled students, who share feature characteristics with both Dropout and Graduate.

Error analysis. The confusion matrix (Figure 7) reveals the dominant error pattern: Graduate students are correctly identified with the highest recall (78.6%), while Enrolled students show the lowest (68.1%), with 12.6% misclassified as Dropout and 19.3% as Graduate. This confirms that Enrolled does not occupy a distinct decision region. The pattern aligns with the t-SNE visualization (Section 2.2) and clustering results (Section 2.3), where Enrolled points lie in the overlapping zone between the two more separable outcomes. The structural overlap between classes, beyond class imbalance alone, is a major source of classification difficulty.

Table 4: Baseline model performance (macro F1 across splits and test accuracy).

Model	Train F1	Val F1	Test F1	Test Acc.
Decision Tree	0.8876	0.6709	0.6752	0.7123
Logistic Reg.	0.7163	0.7076	0.7076	0.7410
SVM (RBF)	0.7752	0.6937	0.6955	0.7319

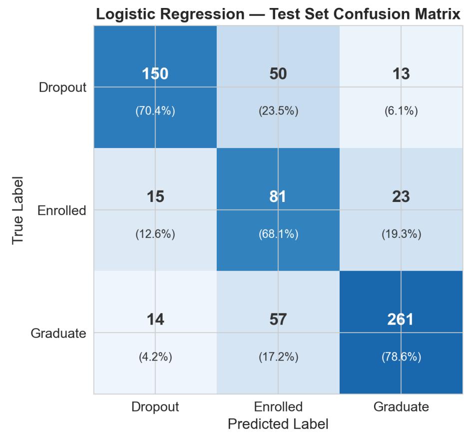

Figure 7: Logistic Regression confusion matrix on the test set.

2.5 Evaluation and Model Choice

We evaluate the three mandatory baselines using held-out precision/recall/F1, One-vs-Rest ROC-AUC, cross-validation, validation curves, learning curves, and GridSearchCV tuning. The overall final model choice is deferred to the open-ended section.

Detailed metrics. Table 5 reports the full set of evaluation metrics on the test set. LR leads among the mandatory baselines by macro F1, while SVM becomes competitive after corrected scaling but still trails LR.

Table 5: Detailed per-model metrics on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)
Decision Tree	0.7123	0.6859	0.6908	0.6752
Logistic Reg.	0.7410	0.7159	0.7237	0.7076
SVM (RBF)	0.7319	0.7054	0.7096	0.6955

Table 6: Per-class and Macro AUC.

Model	Dropout	Enrolled	Graduate	Macro
DT	0.797	0.702	0.849	0.783
LR	**0.910**	**0.822**	0.932	**0.888**
SVM	0.909	0.821	**0.932**	0.887

ROC curves and AUC. Figure 8 compares macro-average ROC curves. LR achieves the highest macro AUC (0.888), closely followed by SVM (0.887) and DT (0.783).

Cross-validation. 5-fold CV on the combined train+val set (3,760 samples) confirms LR as the most stable mandatory baseline: LR CV F1 = 0.7029 ± 0.0064, DT = 0.6511 ± 0.0144, SVM = 0.6955 ± 0.0063.

Validation curve analysis reveals model-specific sensitivity to hyperparameters. Figure 9 shows the two most informative curves: DT max_depth exhibits classic overfitting (training score rises while validation drops), while SVM C shows that increasing regularization flexibility steadily improves both training and validation performance.

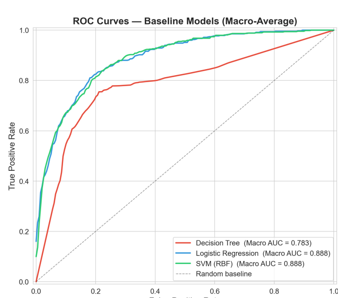

Figure 8: Macro-average ROC curves for all baseline models.

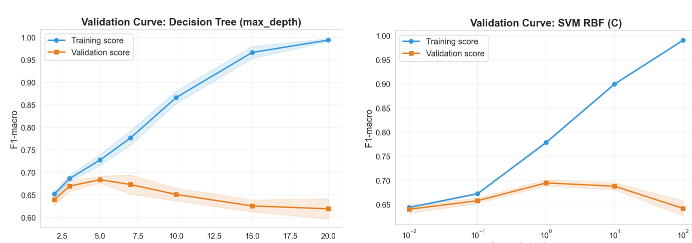

Figure 9: Validation curves: DT max_depth (left) shows overfitting; SVM C (right) shows improvement with larger C.

LR is relatively insensitive to its regularization parameter C, suggesting the linear assumption is the primary bottleneck. SVM gamma (not shown) causes severe overfitting at high values; the optimal C and gamma must be tuned jointly.

Learning curves confirm that DT persistently overfits, LR has nearly converged, and SVM could benefit from more data.

Hyperparameter tuning via GridSearchCV produced the best configurations:

DT:	`max_depth=5, min_samples_split=2, min_samples_leaf=1`	(CV F1=0.6839)
LR:	`C=10, penalty='l2', solver='saga'`	(CV F1=0.7051)
SVM:	`C=10, gamma=0.01`	(CV F1=0.7039)

Table 7: Baseline vs. tuned (F1 macro).

Model	Baseline	Tuned	Δ
DT	0.6752	0.6731	-0.0021
LR	0.7076	0.7074	-0.0002
SVM	0.6955	0.7033	$+0.0078$

SVM benefits most from tuning, improving held-out macro F1 from 0.6955 to 0.7033. DT and LR do not improve on the test set, suggesting that LR's limitation is its linear boundary rather than regularization strength. Enrolled remains the hardest class across all configurations.

Baseline summary. Within the mandatory baselines, LR is strongest and most stable. DT overfits, SVM benefits from tuning but remains below LR, and Enrolled remains the hardest class because it is both underrepresented and structurally intermediate. These findings motivate open-ended exploration beyond baseline deployment.

3 Open-ended Exploration

3.1 Ensemble and Decision-boundary Refinement

The corrected top-40 feature space changes the open-ended question: the baseline bottleneck is no longer simply weak model capacity, but the boundary around the Enrolled class. We therefore evaluated tree ensembles (RF, GB, ExtraTrees) and a validation-tuned LR decision rule. The latter keeps the LR probability model fixed but tunes additive class biases on the validation set to maximize macro F1, then applies the selected biases to the held-out test set.

Table 8: Open-ended model and decision-rule performance on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)	F1 (wtd)
RF (default)	0.7726	0.7163	0.6867	0.6942	0.7600
GradientBoosting (default)	0.7545	0.6883	0.6821	0.6848	0.7516
RF (tuned)	0.7470	0.7120	0.7133	0.7064	0.7554
GradientBoosting (tuned)	**0.7726**	0.7131	0.6965	0.7027	0.7657
ExtraTrees	0.7666	0.7174	0.7141	0.7143	0.7689
LR + class-bias tuning	0.7651	**0.7303**	**0.7290**	**0.7228**	**0.7725**

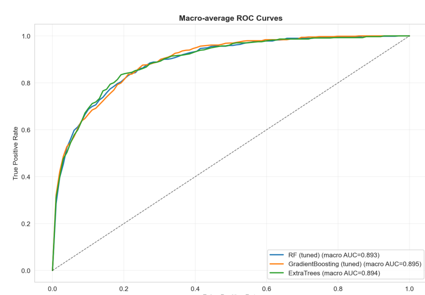

Figure 10: ROC curves for tuned RF, tuned GB, and ExtraTrees.

Ensemble gains are useful but modest: ExtraTrees reaches macro F1 = 0.7143, above RF and GB, while the validation-tuned LR rule gives the largest held-out improvement (0.7228). This supports the interpretation that the remaining error is mainly a class-boundary problem around Enrolled. Since argmax depends only on relative differences, we fix Dropout bias at 0 and search the Enrolled and Graduate offsets. The selected biases [0, 0, +0.4] give Graduate a +0.4 log-odds advantage, correcting the balanced model's tendency to over-predict Enrolled in borderline cases.

3.2 Class Imbalance Handling

Table 9 compares per-class Enrolled F1 with and without balanced class weighting across the three mandatory baseline models after the corrected standardization step.

Table 9: Effect of class weighting on Enrolled F1 score.

Model	Enrolled F1 (default)	Enrolled F1 (balanced)	Δ
Decision Tree	0.4034	0.5016	$+0.0982$
Logistic Reg.	0.4476	0.5277	$+0.0801$
SVM (RBF)	0.4887	0.5180	$+0.0293$

Balanced weighting still improves Enrolled recognition, especially for DT and LR. SVM no longer collapses after scaling correction, so the previous SVM failure was partly preprocessing-related. Remaining Enrolled errors reflect both minority-class pressure and structural overlap.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 11) compared six model classes on the training set:

Table 10: 5-fold CV F1 (macro) results.

Model	CV F1 (Mean $\pm$ Std)	Note
ExtraTrees	0.7170 ± 0.0171	Highest mean
RF	0.7101 ± 0.0208	Strong ensemble
GB	0.7051 ± 0.0093	Stable ensemble
LR	0.7021 ± 0.0079	Strong interpretable baseline
SVM	0.6938 ± 0.0153	Improved after scaling
DT	0.6507 ± 0.0118	Overfitting risk

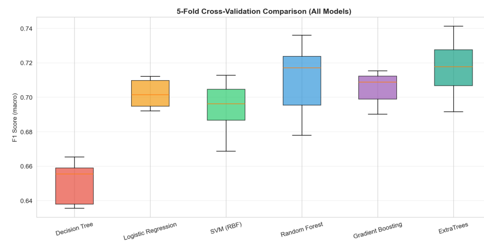

Figure 11: 5-fold CV comparison across model classes.

ExtraTrees has the highest mean CV macro F1, while LR remains close and more interpretable. CV shows that model-family gains are modest; separately, the validation-tuned LR decision rule gives the largest held-out test gain.

3.4 Model Interpretation and Feature Ablation

Feature importance scores from the tuned Random Forest (Figure 12) confirm that academic performance metrics dominate prediction. The top-10 features are led by `overall_approval_rate` and `approval_rate_2nd`, followed by curricular unit metrics and semester grades.

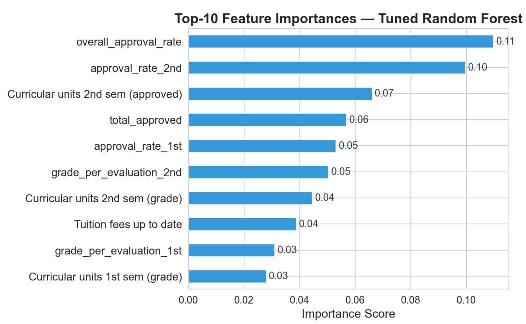

Figure 12: Top-10 feature importances from the tuned Random Forest.

Table 11: Feature group ablation (5-fold CV, tuned RF).

Feature Group	n	CV F1 (Mean $\pm$ Std)
Academic only	20	0.6692 ± 0.0116
Socioeconomic only	20	0.5607 ± 0.0245
All top-40	40	0.7142 ± 0.0073

Feature group ablation. Academic features alone account for most predictive signal (F1 = 0.669), while socioeconomic features alone are weaker (F1 = 0.561). Combining both groups yields the best and most stable performance (F1 = 0.714), confirming complementary information beyond academic records.

Feature count and interaction experiments. Feature count experiments (10/20/30/40 features, 5-fold CV) show that top-40 is best (F1 = 0.700), while top-10 drops to 0.655. Polynomial interactions expanded 40 to 85 dimensions but did not improve CV F1, so we retain the original top-40 space.

3.5 Comprehensive Model Comparison

Table 12 presents all test-set model configurations. Using held-out macro F1 as the criterion, LR with validation-tuned class biases is the final selected decision rule (0.7228), because it directly targets the observed Enrolled boundary problem.

Per-class analysis. Table 13 compares LR+bias with plain LR, both trained on train+validation data. The tuned rule improves Enrolled F1 from 0.528 to 0.538 and Graduate F1 from 0.829 to 0.854 while preserving Dropout performance.

Table 12: Comprehensive comparison of all models on the test set.

Model	Accuracy	F1 (macro)	F1 (weighted)
Decision Tree	0.7123	0.6752	0.7263
Logistic Regression	0.7410	0.7076	0.7550
SVM (RBF)	0.7319	0.6955	0.7448
RF (default)	0.7726	0.6942	0.7600
GradientBoosting (default)	0.7545	0.6848	0.7516
RF (tuned)	0.7470	0.7064	0.7554
GradientBoosting (tuned)	**0.7726**	0.7027	0.7657
ExtraTrees	0.7666	0.7143	0.7689
LR + class-bias tuning	0.7651	**0.7228**	**0.7725**

Table 13: Per-class F1 comparison: LR+bias vs. plain LR.

Class	LR+bias F1	LR F1	Δ
Dropout	0.777	0.774	$+0.004$
Enrolled	0.538	0.528	$+0.010$
Graduate	0.854	0.829	$+0.025$

Statistical significance. Bootstrap analysis (10,000 test-set resamples) gives a 95% CI for LR+bias $-$ LR of $[-0.0026, +0.0278]$ with $p = 0.051$. The interval narrowly crosses zero, so the gain is borderline rather than definitively significant. Still, the direction and error pattern support targeted decision-boundary refinement.

4 Conclusion

Key findings: The engineered top-40 feature space contains strong predictive signal, with academic metrics providing the primary contribution (academic-only F1 = 0.669; all top-40 F1 = 0.714). t-SNE and clustering show that the outcomes are not naturally separated because Enrolled students lie between Dropout and Graduate. Among mandatory baselines, LR is strongest and most stable; DT overfits and SVM improves once the final top-40 space is correctly standardized. In open-ended exploration, LR with validation-tuned class biases achieves the best held-out test macro F1 of 0.7228 versus 0.7102 for plain LR, though the bootstrap result is borderline ($p = 0.051$). Enrolled remains the primary challenge because it is both a minority class and a structurally transitional outcome.

Limitations: The borderline gap between LR+bias and plain LR means the final result should not be treated as a definitive deployment advantage without external validation. The t-SNE decision-boundary plots are 2D approximations, and clustering struggled because discriminative features do not correspond to natural density-based clusters.

Future work: Future work should focus on better modeling of Enrolled students, external validation, and temporal academic trajectories for early-warning deployment.

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[3] Van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. JMLR, 9, pp. 2579–2605.

[4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Member	Contribution
Hao LUO	[Data preprocessing, feature engineering, report writing]
Congren HANG	[Clustering analysis, t-SNE visualization, report writing]
Xingjian LI	[Model training, evaluation, report writing]
Ruoxi WANG	[Open-ended exploration, presentation]